

作业

11.21 Consider an op amp having a single-pole, open-loop response with $A_0 = 10^5$ and $f_p = 10$ Hz. Let the op amp be ideal otherwise (infinite input impedance, zero output impedance, etc.). If this amplifier is connected in the noninverting configuration with a nominal low-frequency, closed-loop gain of 100, find the frequency at which $|A\beta| = 1$. Also, find the phase margin.

Ans. 10^4 Hz; 90°

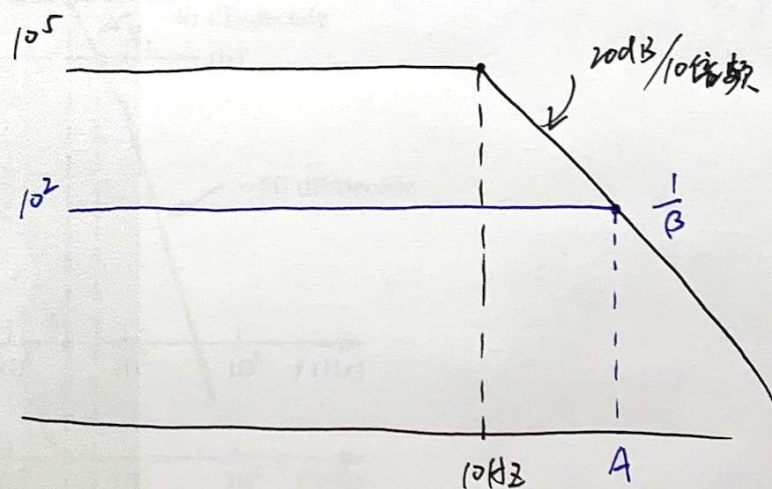
① $\frac{1}{\beta} = 100$, 为图中蓝色线

$\therefore A$ 点频率为

$$10 \text{ Hz} \times 10^3 = 10^4 \text{ Hz}$$

② 单极点, $\therefore A$ 点相位为 -90°

$$PM = 90^\circ$$



11.23 For the amplifier whose open-loop-gain frequency response is shown in Fig. 11.38, find the value of β that results in a phase margin of 45° . What is the corresponding closed-loop gain?

Ans. 10^{-4} ; 80 dB

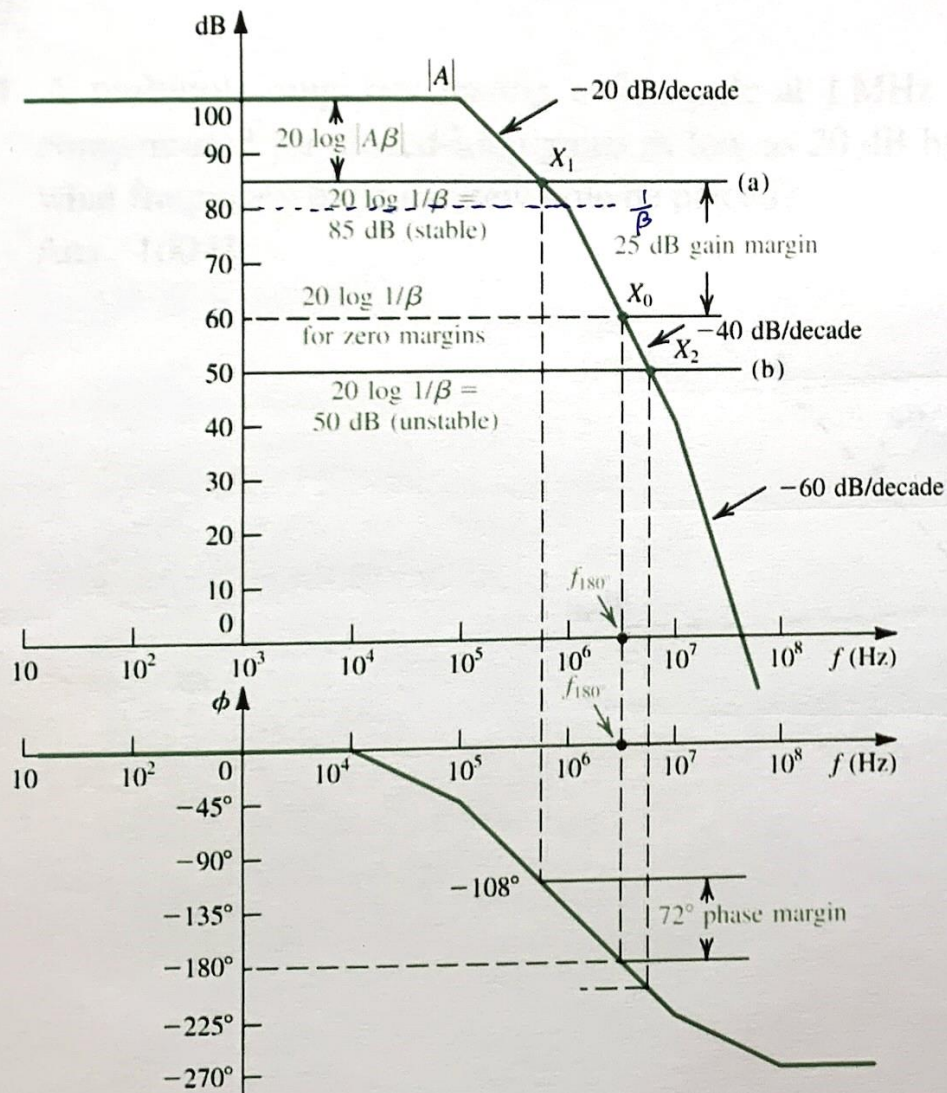
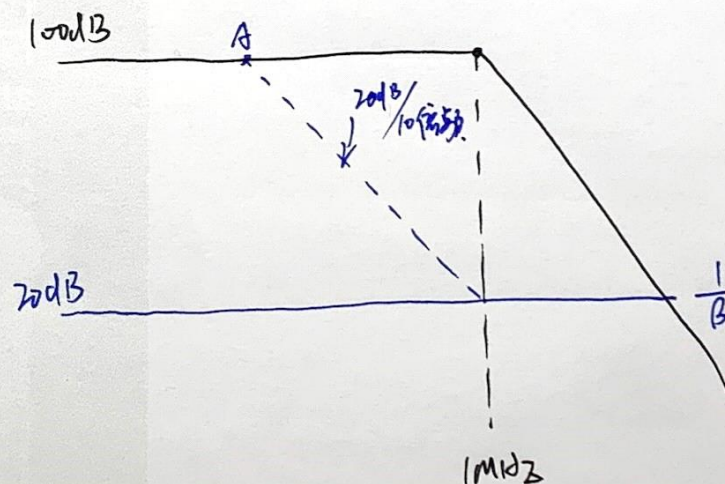


Figure 11.38 Stability analysis using Bode plot of A .

- 11.24 A multipole amplifier having a first pole at 1 MHz and a dc open-loop gain of 100 dB is to be compensated for closed-loop gains as low as 20 dB by the introduction of a new dominant pole. At what frequency must the new pole be placed?

Ans. 100 Hz



$$A \text{ 点频率} = \frac{1 \text{ MHz}}{10^4} = 100 \text{ Hz}$$